Evaluating Centralized Value Learning in Shared-Policy MARL for Energy-Aware Multi-Site 5G UPF Selection

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***Abstract*—Energy-aware User Plane Function (UPF) selection at the 5G edge must balance low-load energy savings against quality-of-service (QoS) degradation as traffic changes. We compare shared-policy controllers trained with centralized and local value estimation across heterogeneous edge sites. A measurement-calibrated simulator driven by real mobile-traffic traces models DPDK and user-space UPF energy, QoS, and activation costs. We train MAPPO with a fleet-state critic and compare it with Shared-PPO using a local critic under common actor and training settings, alongside independent PPO, centralized PPO, optimized hysteresis, and always-DPDK. MAPPO achieves the lowest mean combined cost, reducing it by 10.2% relative to hysteresis and 21.9% relative to always-DPDK, with energy reductions of 1.4% and 5.0%, respectively. Against Shared-PPO, MAPPO lowers mean combined cost by 1.45% and reduces seed-to-seed variability. Despite 98.5% decision agreement, MAPPO more often avoids user-space assignments that are currently safe but become costly after a demand rise during transition. Calibration and reward-sensitivity analyses show robust gains over independent learning and hysteresis, while the smaller shared-policy gap depends on the modeled user-space QoS boundary.**

***Index Terms*—5G Core, User Plane Function, energy-aware orchestration, multi-agent reinforcement learning, centralized training with decentralized execution.**

I. Introduction

Distributing the User Plane Function (UPF) toward the 5G network edge enables local traffic steering and reduces the distance between users and services, but also increases the number of continuously provisioned packet-processing instances. Their energy efficiency depends on both the offered traffic and the implementation used to process it. A DPDK data plane sustains high loads through polling and kernel bypass, whereas a user-space implementation can consume less power during light traffic but reach its performance limits earlier [1], [2]. Consequently, using one implementation throughout the day sacrifices either low-load efficiency or service quality, while switching between implementations introduces activation costs that a controller must weigh against the expected saving.

This decision becomes more difficult across sites with heterogeneous traffic patterns, because the value of a low-power assignment depends on how long the favorable conditions persist. Independent policies learn this trade-off from their own sites, even when relevant switching conditions occur infrequently. A shared policy can instead learn from the collective experience of the fleet while selecting each site's UPF from local observations. Under centralized training with decentralized execution (CTDE), a critic can use observations from all sites to estimate future returns during training, although the deployed actor requires only local information. Whether this additional training structure improves the energy–QoS trade-off beyond policy sharing itself remains an empirical question.

We investigate this question through a simulation environment calibrated from physical UPF profiling and driven by real mobile-traffic traces. The profiles provide the power and performance response of each implementation, allowing controllers to experience different traffic and switching sequences without repeating a physical measurement for every policy. Our study compares MAPPO with a shared-actor PPO controller using a local critic, alongside independent PPO, centralized PPO, and an optimized hysteresis baseline. The shared-actor comparison fixes the actor and common PPO settings, while critic information, capacity, and output structure differ.

The contributions are threefold. First, we formulate multi-site UPF selection with explicit QoS, activation-energy, and anti-oscillation costs. Second, an eight-seed comparison and analysis of saved decisions explain how the learned policies balance energy savings against QoS. Third, ten alternative calibrations and 81 reward-weight combinations establish which conclusions remain stable when the fitted profiles and operating priorities change. Together, these analyses connect aggregate performance to the conditions under which shared-policy control is useful.

II. Related Work

UPF optimization operates at several decision levels. Leyva-Pupo and Cervelló-Pastor [3] formulate placement and chaining as a multi-objective integer program that balances provisioning expenditure and QoS, and develop heuristics for larger instances. Their decision concerns where UPFs and service chains should be deployed. Our deployment locations are fixed, while the controller repeatedly selects the active implementation at each site as its traffic changes. Measurement studies by Christakis *et al.* [1] and Bolla *et al.* [2] characterize the performance and energy profiles that motivate such selection, but do not address how a policy should learn its temporal trade-off.

Bellin *et al.* [4] apply deep reinforcement learning to assign new sessions to distributed edge or cloud UPFs using power, latency, and bandwidth information. Their orchestrator selects

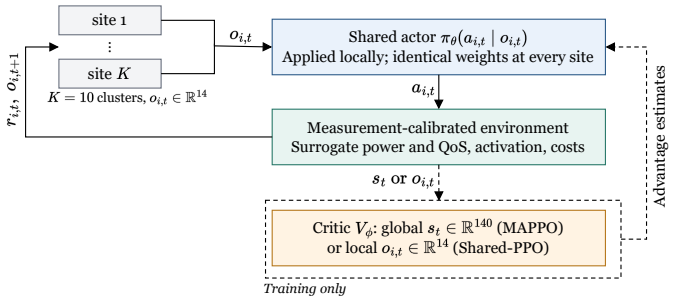


Fig. 1. Shared-actor control loop. Solid arrows show local execution at each of ten sites; dashed arrows show training through a global-state critic (MAPPO) or a shared local critic (Shared-PPO).

an anchor when a connection is established, whereas we study repeated implementation changes at existing sites, for which the previous UPF and activation overhead affect the next decision. Our earlier work [5] combines measured UPF profiles with LSTM traffic predictions, a utility-based selection rule, and hysteresis and hold-down mechanisms. The present work extends the control setting to heterogeneous sites and learned shared policies, using an optimized version of the threshold-based strategy as a reference.

The learning architectures build on PPO [6] and its multi-agent extensions. Yu *et al.* [7] show that MAPPO can be effective in cooperative tasks through CTDE, while Schroeder de Witt *et al.* [8] demonstrate that locally estimated value functions can also support competitive PPO policies on the StarCraft multi-agent benchmark. Thus, a centralized critic is not automatically preferable, and the value of additional training information must be distinguished from that of sharing experience across different traffic distributions. We compare the critics with the same shared actor and common training settings, without isolating global information from critic capacity and output structure, and assess their energy–QoS behavior and calibration sensitivity.

III. System and Control Model

A. Fleet, traffic, and decision loop

We consider $K = 10$ multi-access edge computing (MEC) clusters. At each 15-minute step, cluster i selects $a_{i,t} \in \{0, 1\}$, where 0 denotes the SD-Core DPDK UPF [9] and 1 the OpenAirInterface user-space UPF [10]. The Session Management Function (SMF) and the remaining 5G Core control-plane functions remain centralized; only the active data-plane realization changes. The environment follows the PettingZoo parallel interface [11]. There is no fleet-wide power or capacity constraint, and one site's action does not affect another site's traffic or costs. Sites therefore couple only through shared training.

Traffic comes from the NetMob23 Lyon dataset [12]. Grid demand is mapped to radio-site service areas and aggregated into ten clusters with different load distributions. The chronological train, validation, and test slices contain 5073, 1009, and 1009 steps, corresponding to approximately 53, 10.5, and 10.5 days. The saved forecasts come from a cluster-first Graph WaveNet model [13], which combines graph convolution over the inter-cluster graph with dilated temporal convolutions to capture spatial and temporal traffic dependencies. These predictions provide the forecast feature used by all controllers. Each decentralized actor receives a 14-dimensional local observation comprising current load, eight past loads, forecast, previous UPF, previous QoS score, previous specific energy, and cooldown progress. Mean test loads range from approximately 0.1 to 1.7 Gbps, covering sites with frequent low-load opportunities and sites operating predominantly above the user-space QoS range.

B. Profiling-based simulation environment

The simulation environment is calibrated from 220 constant-load profiling runs, organized as five chronological sweeps of 22 offered-load levels for each UPF. Each sweep repeats the same levels, spanning approximately 0–5 Gbps, so the dataset captures both repeated measurements at a given operating point and differences across traffic loads. Keysight LoadCore [14] supplies throughput, delay, and packet-loss measurements, which are aligned with process-attributed power series from Scaphandre [15], using RAPL measurements [16].

The fitted models form two layers for each UPF. Layer 1 predicts throughput, CPU utilization, delay, and loss from offered traffic, using ridge regression, gradient-boosted trees, or random forests according to the selected target model. Layer 2 predicts power from offered traffic and the Layer 1 outputs using a random forest. DPDK loss uses a constant model where no loss was observed in profiling. This structure links energy consumption to the predicted operating condition rather than assigning a fixed power value to each UPF.

The simulator combines these predictors with traffic traces and measured activation parameters. At each decision, it evaluates the selected implementation, accounts for the previous UPF during a transition, and returns the resulting reward and observation. The repeated-sweep and load-level structure also supports the calibration tests in Sec. IV-C, which refit the models after withholding complete measurement groups.

C. QoS score and specific energy

Let $d_{i,t}$ and $\ell_{i,t}$ be the predicted delay and lost-packet count for the selected realization. With $[x]_+ = \max(0, x)$, the normalized scores are

$$q^d_{i,t} = \left[1 - \frac{[d_{i,t} - d_{\max}]_+}{d_{\max}}\right]_+, \tag{1}$$

$$q^\ell_{i,t} = \left[1 - \frac{[\ell_{i,t} - \ell_{\max}]_+}{\ell_{\max}}\right]_+, \tag{2}$$

$$Q_{i,t} = \min\{q^d_{i,t}, q^\ell_{i,t}\}. \tag{3}$$

Here $d_{\max} = 200$ μs and $\ell_{\max} = 5$ packets per 3-s measurement interval. Each score equals one within its budget and decreases with relative excess, reaching zero at twice the budget. Taking the minimum prevents good delay from compensating for excessive loss, or vice versa. The reward

target $\tau = 0.9$ penalizes score shortfall; the reported QoS-violation rate instead counts any budget exceedance. One cluster-step is one site's 15-minute decision, so QoSV% is the percentage of decisions violating a budget, not a packet-loss percentage. Score shortfall and budget exceedance need not coincide for small excesses.

Specific energy consumption (SEC) normalizes steady-state operating power by demand,

$$\mathrm{SEC}_{i,t} = \frac{P^{op}_{i,t}}{\max(\lambda_{i,t}, \epsilon)}. \tag{4}$$

Power is in watts, offered load $\lambda_{i,t}$ in Mbps, and $\epsilon > 0$ guards against division by zero. Thus SEC has units J/Mbit; it is energy per offered traffic volume, not per successfully delivered bit. A nearly constant DPDK power draw gives a large SEC at light load, making low-load realization selection valuable even when its contribution to total fleet watt-hours is small. The objective sums site costs without traffic weighting, so minimizing SEC and minimizing fleet energy are distinct criteria.

D. Objective and transition costs

The per-site reward is the negative of the combined cost, comprising four non-negative terms,

$$r_{i,t} = -\left[\alpha \mathrm{SEC}_{i,t} + \lambda_q[\tau - Q_{i,t}]_+ + \lambda_s E^s_{i,t} + C^c_{i,t}\right], \tag{5}$$

where $\alpha = 100$, $\lambda_q = 30$, $\tau = 0.9$, and $\lambda_s = 4$. Reward is summed over sites and test steps; higher is better. Energy, QoS violations, user-space share, and switching rate are also reported separately.

Let $\chi_{i,t}$ indicate a realized change and $e^{act}(a_{i,t})$ denote the activation energy of the new realization. The switching term is $E^s_{i,t} = \chi_{i,t} e^{act}(a_{i,t})$. Activation takes 24 s and 0.009476 Wh for DPDK, versus 3.3 s and 5.15×10^{-6} Wh for the user-space UPF. The energy report uses sub-step accounting. During activation the old UPF serves traffic; the new UPF serves the remainder of the $\Delta = 900$ s step. Writing $u_{i,t} = \chi_{i,t} t^{act}(a_{i,t})$, the step energy in Wh is

$$E_{i,t} = \frac{u_{i,t} P^{old}_{i,t} + (\Delta - u_{i,t}) P^{new}_{i,t}}{3600} + E^s_{i,t}. \tag{6}$$

Fleet energy sums $E_{i,t}$, counting activation energy once, whereas $\lambda_s E^s_{i,t}$ is its weighted reward charge. SEC uses steady-state power. Switching-step QoS uses the larger predicted loss of the old and new UPFs at the current load, rather than a duration-weighted loss. A switch to DPDK can therefore incur a QoS penalty if the old user-space UPF is unsafe during activation.

The four-step anti-oscillation window adds a soft dwell cost. If $h_{i,t}$ is the number of steps remaining in that window before a change, then

$$C^c_{i,t} = 0.5\, \chi_{i,t} \frac{h_{i,t}}{4}, \qquad 0 \le h_{i,t} \le 4. \tag{7}$$

The charge decreases as the window expires and is zero after four steps (one hour). It discourages rapid reversals while allowing a QoS-driven switch; it is neither an additional energy measurement nor a hard action prohibition. Cooldown progress and the previous realization are included in the local observation. Reported switches per cluster-day count consecutive-step changes; a first-step change from the initial DPDK state incurs transition cost but is not included in that switching count.

TABLE I
ARCHITECTURE AND COMMON TRAINING SETTINGS. NETWORK ENTRIES GIVE LAYER WIDTHS.

Setting	MAPPO	Shared-PPO
Shared actor	14–64–64–2 (5250 parameters)	
Critic	140–128–128–10	14–128–128–1
Actor initialization	Same seed; actor constructed first	
Training budget	200,000 environment steps	
Rollout / minibatch	1024 steps / 256 samples	
Epochs / PPO clip	10 / 0.2	
Discount / GAE decay	$\gamma = 0.995$ / $\lambda = 0.9$	
Learning rate / entropy	3×10^{-4} / 0.05	
Joint actor–critic gradient clip	0.5 (norm limit)	
Checkpoint / test action	Best validation reward / argmax	

The cooperative training objective is $J(\theta) = \mathbb{E}_{\pi_\theta}[\sum_t \gamma^t \sum_{i=1}^K r_{i,t}]$. Fig. 1 shows the shared actor $\pi_\theta(a_{i,t} \mid o_{i,t})$ executed locally. MAPPO's critic predicts site returns from the concatenated fleet state; Shared-PPO uses a shared local critic. Training pools site–time generalized-advantage estimates for actor updates. Execution requires no critic, global state, or inter-site communication.

IV. Controllers and Evaluation Protocol

A. Controller architectures

MAPPO's critic has 35,850 parameters and one output per cluster; Shared-PPO's shared local critic has 18,561 parameters and one output without a cluster identifier. Thus the comparison changes information, capacity, and value-head structure together.

IPPO trains one Stable-Baselines3 PPO policy per site [17]. Centralized PPO maps the 140-dimensional fleet state to a $\mathrm{MultiDiscrete}([2]^K)$ joint action. These are practical system baselines, whereas the MAPPO–Shared-PPO pair compares value architectures within a common shared-actor design. In particular, IPPO retains its independent trainer and centralized PPO retains its load-weighted joint training signal. Every controller is evaluated with (5).

B. Training and reference controllers

Table I summarizes the architecture and training settings, including the common PPO and generalized-advantage parameters [6], [18]. All learned controllers use the same environment version and eight seeds $\{1, 7, 13, 23, 42, 64, 77, 99\}$, with the best-validation checkpoint evaluated deterministically on the complete test slice.

Always-DPDK provides a static reference, while fixed hysteresis is an optimized adaptation of our earlier forecast-driven threshold strategy [5]. It switches to DPDK above 81 Mbps and returns to user space below 31 Mbps, retaining the current UPF between these thresholds to suppress oscillation.

The 50-Mbps band is selected on the saved test trace, making this an optimistic rather than held-out reference; it remains fixed during calibration tests.

Results retain all seeds and report mean reward, sample standard deviation, and median. Paired differences use 10,000 bootstrap resamples for 95% confidence intervals (CIs) and an exact two-sided sign-flip test. Percentages are calculated from unrounded means.

C. Sensitivity to simulator calibration

Since the environment is fitted from a finite profiling campaign, the controller comparison could in principle reflect the surrogate fit rather than the policies; we therefore refit the models after withholding complete measurement groups. *Leave-one-sweep-out (LOSO)* omits one of the five sweeps and refits on the other four, testing measurement repetition while retaining every load level. *Leave-one-load-group-out (LOLO)* sorts the 22 levels into five interleaved groups by rank modulo five and omits one group across all sweeps, testing predictions across gaps in measured load coverage. The resulting ten environments retain the model configuration, controller weights, traffic, and forecasts.

In *closed-policy evaluation*, the saved policy interacts with each refit, allowing changed observations to affect decisions. *Fixed-action replay* instead imposes its original action sequence and recomputes the outcomes. Comparing these modes separates changes in decisions from changes in their assessed cost. The safe-range endpoint in Fig. 4(b) is the last load on a 1-Mbps grid before the fitted model first predicts a delay or loss budget violation, a property of the surrogate rather than a measured capacity.

V. Results

A. Overall performance and consistency

MAPPO achieves the lowest mean combined cost while preserving most of the energy benefit of dynamic UPF selection (Table II). Relative to fixed hysteresis, it reduces combined cost by 10.2% and energy by 1.4%. Against always-DPDK, the corresponding reductions are 21.9% and 5.0%, with the rounded table totals showing a 104-Wh saving over the test period. Always-DPDK has fewer QoS violations, whereas Shared-PPO uses slightly less energy than MAPPO but incurs more violations. MAPPO therefore offers the best balance under the stated objective, rather than minimizing energy or QoS violations in isolation.

The shared policies also perform more consistently across seeds (Fig. 2). IPPO's mean is affected by seed 99's reward of -25341, whereas its median of -5794 shows that most runs are much closer to the shared policies. Centralized PPO varies widely and remains worse than always-DPDK in every seed, indicating that access to the fleet state alone does not ensure effective control.

B. How the shared policies balance energy and QoS

Under identical reward weights, Shared-PPO favors greater user-space use and lower energy consumption, whereas MAPPO accepts additional energy cost to reduce QoS penalties. MAPPO's mean combined cost is 1.45% lower, with positive reward differences in seven of eight pairs and standard deviation reduced from 77 to 22 (Table III). The evidence is strongest for combined reward, since the energy CI includes zero and the exact QoS-cost test gives $p = 0.102$.

The load-dependent choices in Fig. 3(a) locate this trade-off. Below 25 Mbps both policies almost always select user space, where its energy advantage is pronounced. In the 50–81 Mbps band, Shared-PPO selects it in 27.3% of steps compared with MAPPO's 18.4%, despite agreement on 98.5% of all cluster-steps. All 876 additional user-space assignments made only by Shared-PPO are safe at that moment, so their consequence cannot be understood from current-load feasibility alone. The saved sequences show that a rise in demand can expose the old user-space UPF to unsafe load while DPDK starts. MAPPO's more conservative choices reduce this exposure at the expense of some immediate savings, with 14.1 penalized transitions per test episode against Shared-PPO's 19.1. Overall, its QoS cost is 106.9 units lower, offset by 25.4 additional SEC-cost units, while activation and cooldown contribute a further 1.4 units to its net gain (Fig. 3(b)).

TABLE II
Test results over ten sites and 1009 steps. Reward gives mean ± standard deviation and median across eight seeds. Other columns give mean energy (whole Wh), QoS-violating cluster-steps, user-space use, and switches per cluster-day.

Controller	Reward	Median	Wh	QoSV%	USR%	Sw./d
MAPPO	$\mathbf{-5624 \pm 22}$	-5626	1967	0.52	7.8	2.88
Shared-PPO	-5707 ± 77	-5704	1963	0.57	8.5	3.00
IPPO	-8236 ± 6912	-5794	2026	1.47	8.9	2.03
Central PPO	-15025 ± 6931	-13365	2132	4.03	11.1	1.66
Hysteresis	-6262	—	1995	0.68	7.5	1.06
Always-DPDK	-7201	—	2071	0.38	0.0	0.00

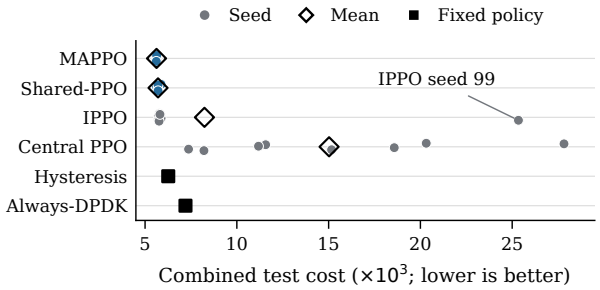


Fig. 2. Combined test cost (negative reward) across training seeds. Dots show all eight seeds, diamonds their means, and squares fixed policies.

TABLE III
Paired differences across eight seeds, MAPPO minus Shared-PPO.

Metric	Mean difference	95% CI	Exact p
Reward	$+82.9$	$[31.0, 150.9]$	0.023
QoS cost	-106.9	$[-203.4, -9.4]$	0.102
Energy (Wh)	$+4.2$	$[-2.7, 10.3]$	0.289

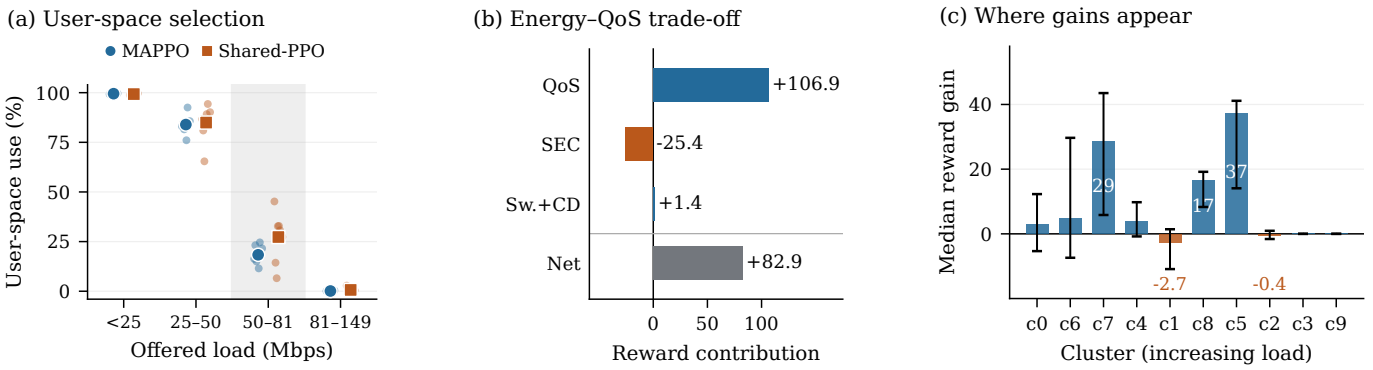


Fig. 3. Energy–QoS behavior of the shared policies. (a) User-space use by load band, with individual seeds and means. (b) QoS and SEC contributions, combined activation/cooldown contribution (Sw.+CD), and net reward gain. (c) Median reward gain by cluster, ordered by mean load, with whiskers spanning the middle 50% of seed results. Orange marks negative medians.

The cluster comparison distinguishes sustained heavy traffic from mixed-load operation. At c3 and c9, user-space opportunities are rare, leaving little scope for the policies to differ. At c5, c7, and c8, low-load intervals recur within otherwise higher-load operation, and avoiding assignments before a load increase produces the largest gains (Fig. 3(c)). Cluster c1 also has low-load opportunities, but fewer direct jumps from below 81 to above 149 Mbps per opportunity than c5, c7, and c8. MAPPO's caution therefore recovers too little QoS cost to offset the saving lost, yielding a median gain of -2.74; c2 is also slightly negative (-0.44). The sequence of operating conditions, rather than mean load alone, determines the benefit.

C. What calibration changes reveal

MAPPO beats IPPO and hysteresis in all ten refits, ranking first among all six controllers in eight (Fig. 4). All five LOSO refits favor MAPPO, whereas LOLO groups L2 and L3 favor always-DPDK and Shared-PPO, respectively. The comparison is therefore less sensitive to repeated sweeps than to missing loads near the user-space performance transition.

L2 omits 1, 10, 100, and 1000 Mbps, removing the measured 100-Mbps operating point near the user-space QoS transition. All 200 withheld user-space samples at that level satisfy the QoS budgets, but the refitted model predicts them unsafe. Its first predicted violation therefore occurs at a lower load, with the safe-range endpoint shifting from 149 to 81 Mbps. MAPPO still uses less energy than always-DPDK (1982 versus 2068 Wh), but its QoS cost rises to 3329 against 1024, outweighing that saving. The reversal reflects a pessimistic prediction of user-space performance, not a physical change in the UPF.

L3 omits 2, 20, 200, and 2000 Mbps and shifts the endpoint upward to 170 Mbps, making user-space operation less costly and giving Shared-PPO a 27.3-unit reward advantage. Yet MAPPO's actions change by at most 0.019% under LOSO and 0.171% under LOLO, and the winning controller is unchanged between closed-policy evaluation and fixed-action replay in every refit. Recalibration therefore mainly changes the estimated consequences of similar decisions. The advantage

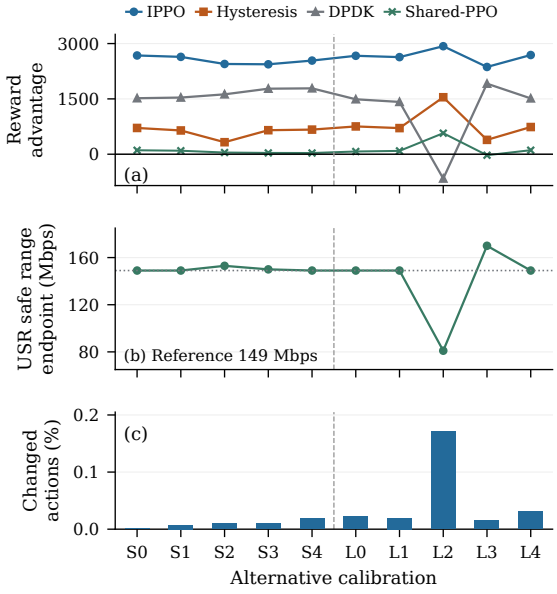


Fig. 4. Calibration sensitivity across five omitted sweeps (S0–S4) and five omitted load groups (L0–L4). (a) MAPPO's mean reward advantage. (b) Predicted onset of user-space QoS violations, expressed as offered load. (c) MAPPO actions differing from the reference calibration. Policies act in each refitted environment without retraining.

over IPPO and hysteresis survives these changes, whereas the closer energy–QoS choice between the shared policies depends on accurately representing the user-space performance transition.

D. Sensitivity to reward priorities

Rescoring the saved decisions with each of the four cost weights multiplied by 0.5, 1, or 2 gives $3^4 = 81$ combinations. MAPPO ranks first among all six controllers throughout, with mean reward margins of 3.3–203.6 units (Fig. 5). The smallest margin occurs when SEC receives twice its original weight and the other costs receive half, making Shared-PPO's energy-saving behavior more attractive. Thus MAPPO's advantage is not confined to the original weighting, although it narrows when

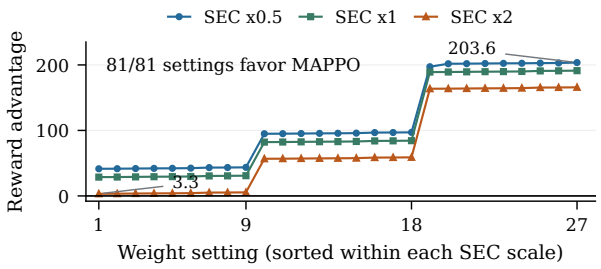


Fig. 5. Ranking sensitivity using saved decisions from all six controllers. Each curve independently sorts the 27 QoS, switching, and cooldown weight combinations for one SEC multiplier. All 81 reward margins are positive.

energy efficiency takes greater priority. This is a sensitivity test of the recorded policies, not an evaluation of policies retrained under different objectives.

VI. Discussion

The results favor shared local control that values low-load energy opportunities over their expected duration. Capturing every immediate saving can increase later costs when demand changes, while excessive caution forfeits savings where those changes are less consequential. This explains the different energy–QoS balances and the concentration of MAPPO's gains at mixed-load sites.

The centralized critic uses fleet context and separate site returns, whereas the local critic fits one value function across sites. Critic capacity, output structure, and checkpoint timing also differ, preventing attribution to global information alone. SEC divides power by load, so the 684 validation cluster-steps below 1 Mbps, versus four in training and none in test, dominate checkpoint selection; the small MAPPO–Shared-PPO gap may therefore partly reflect validation mismatch.

For transfer, calibration should cover the user-space performance transition. The simulator uses conservative, non-duration-weighted activation loss, so transition-related QoS gains depend on this modeling choice; energy profiles are process-attributed. These refits vary the fit of one campaign, not the hardware or traffic region. The centralized critic adds value-estimation complexity only during training, leaving the deployed shared actor and its local execution unchanged.

VII. Conclusion

Shared-policy learning combines experience from heterogeneous UPF sites while preserving local execution. MAPPO's observed advantage is associated with a learned energy–QoS trade-off, but cannot be attributed to global critic information alone because critic capacity and output structure also differ. Because this balance depends on subsequent traffic and the fitted UPF profiles, deployment should prioritize validation around the user-space performance transition.

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